PRODUCT USER MANUAL

OMI_AMOC
Atlantic meridional Overturning Circulation
GLOBAL_OMI_amoc_max26N_timeseries
GLOBAL_OMI_amoc_26N_profile

Issue 3.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Checked by	Validated by
1.0	2018/06/05	All	Creation of the document	Laura Jackson		
2.0	2022/04/04	All	Update of template	Flora Gues		Marie Drevillon
3.0	2024/06/30	All	Update with new product and extension year 2023	Olivier Le Galloudec		Marie Drévillon

GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document
AMOC	Atlantic Meridional Overturning Circulation
CGLOR	C-GLORSv7 reanalysis
CMEMS	Copernicus Marine Environment Monitoring Service
GLO	Global
GLORYS	GLORYS2V4 reanalysis
ORAS5	ORAS5 reanalysis
GREP	Global Reanalysis Ensemble Product

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DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I INTRODUCTION

I.1. Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) OMI_AMOC : it provides time series of AMOC strength at 26.5°N, and time mean profiles of the AMOC at 26.5°N. See QUID document for more details of calculation.
OMI_AMOC has two datasets :

- GLOBAL_OMI_amoc_max26N_timeseries which contains monthly mean time series of the AMOC strength calculated as the maximum in depth of each monthly profile at 26.5°N from 1993-2023
- GLOBAL_OMI_amoc_26N_profile which contains time mean (1993-2023) profiles of the AMOC at 26.5°N. Profiles are 1D (depth).

These data are computed using GLORYS, CGLOR and ORAS reanalysis (GREP[1]).

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

Products GLOBAL_OMI_amoc_max26N_timeseries and GLOBAL_OMI_amoc_26N_profile are described in the Quality Information Document (QUID) CMEMS-QUID-OMI-amoc.pdf (CMEMS-QUID-OMI_amoc.pdf (copernicus.eu)).

[1] GLOBAL_MULTIYEAR_PHY_ENS_001_031

I.2. History of changes

2018/07	1st publication of OMI_AMOC (GLOBAL_OMI_NATLANTIC_amoc_26n_profile; GLOBAL_OMI_NATLANTIC_amoc_max26n_timeseries)
2022/04	Update of template
2024/07	Update product GREP and extension

II DESCRIPTION OF THE PRODUCT

II.1. General information

Product(s) name	GLOBAL_OMI_amoc_max26N_timeseries ; GLOBAL_OMI_amoc_26N_profile
Geographical coverage	26.5°N
Variables	Atlantic Meridional Overturning Circulation (amoc)
Update frequency	Annual
Available time series	1993-2023
Delivery mechanism	Marine Data Store
Temporal/Spatial resolution	Monthly averages
Integration depth	Full depth
Format	NetCDF CF1.7

II.2. Details of datasets

GLOBAL_OMI_amoc_max26N_timeseries contains the 1D monthly AMOC timeseries at 26.5°N: from CGLORS, GLORYS, GLOSEA, ORAS and the ensemble mean and standard deviation AMOC.
amoc_mean [m3/s] Ensemble Mean of Atlantic Meridional Overturning Circulation strength maximum_of_amoc_streamfunction_at_26.5N **amoc_std [m3/s]** Ensemble Standard Deviation of Atlantic Meridional Overturning Circulation strength maximum_of_amoc_streamfunction_at_26.5N **amoc_glor [m3/s]** Atlantic Meridional Overturning Circulation strength from GLORYS maximum_of_amoc_streamfunction_at_26.5N **amoc_cglo [m3/s]** Atlantic Meridional Overturning Circulation strength from CGLORS maximum_of_amoc_streamfunction_at_26.5N **amoc_oras [m3/s]** Atlantic Meridional Overturning Circulation strength from ORAS maximum_of_amoc_streamfunction_at_26.5N
GLOBAL_OMI_amoc_26N_profile contains the 1D time mean AMOC profile at 26.5°N: from CGLORS, GLORYS, ORAS and the ensemble mean and standard deviation AMOC.
amoc_mean [m3/s] Ensemble Mean of Atlantic Meridional Overturning Circulation profile profile_of_amoc_streamfunction_at_26.5N **amoc_std [m3/s]** Ensemble Standard Deviation of Atlantic Meridional Overturning Circulation profile profile_of_amoc_streamfunction_at_26.5N **amoc_glor [m3/s]** Atlantic Meridional Overturning Circulation profile from GLORYS profile_of_amoc_streamfunction_at_26.5N **amoc_cglo [m3/s]** Atlantic Meridional Overturning Circulation profile from CGLORS

profile_of_amoc_streamfunction_at_26.5N
amoc_oras [m3/s]
Atlantic Meridional Overturning Circulation profile from ORAS
profile_of_amoc_streamfunction_at_26.5N

II.3. Product System Description

II.3.1. GLOBAL_OMI_amoc_max26N_timeseries

Data extracted from GREP provides monthly mean velocities which are used to calculate profiles of the AMOC at 26.5°N. The monthly time series of AMOC strength is calculated as the maximum in depth of these profiles. It starts in January 1993 and finishes in December 2023.

II.3.2. GLOBAL_OMI_amoc_26N_profile

Data extracted from GREP provides monthly mean velocities which are used to calculate profiles of the AMOC at 26.5°N. The time mean profile is calculated over 1993-2023 and is given as a function of depth.

III FILE FORMAT

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1. Nomenclature of files

OMI_AMOC files nomenclature when downloaded through the CMEMS MDS Services is based on the following nomenclature:

<datasetname>_<timestep>_P<productiontime>.nc

Where :

- **<datasetname>** is the dataset reference **(**global_omi_amoc_max26n_timeseries or global_omi_amoc_26n_profile).
- **<timestep>** is the first time step contained in the file**.**
- **<productiontime>** is the production time when the file was generated.
- **.nc** is the standard NetCDF filename extension.

Example: global_omi_amoc_26n_profile_19930115_P20180601.nc

V REFERENCES

See the Quality Information Document (QUID) and Product User Manual (PUM): CMEMS-QUID-OMI_amoc.pdf (copernicus.eu)